Jack Hogan

703-919-2976 jackhogan11@gmail.com www.jackhogan.me gh/ImTheSquid in/jackhogan11

Computer science and artificial intelligence engineer with experience across software engineering, AI/ML research, and technical leadership. Focused on deep reinforcement learning and systems that let models reason, adapt, and learn complex behaviors, with applications in robotics, vision, and low-data learning.

Experience

Era Labs
Founding Embedded Systems Engineer
August 2026 — Present San Francisco, CA

- Designed and implemented a patent-pending firmware architecture supporting dynamic peripheral registration and bytecode execution for ~100ms idea iteration.
- Designing and fabricating circuits for new product lines with microamp-scale power draws and efficient layouts.
- Designed a cohesive software stack for registering, finding, and collaborating with Beacons, a hardware/software solution for fostering real-time social connections at events, using traditional and LLM-based machine learning techniques.

Peraton Labs
Machine Learning Researcher
May 2025 — May 2026 Silver Spring, MD

- Created a new pipeline for concurrently testing different GNN architectures, speeding up testing by 2-3x.
- Investigated and implemented multiple bleeding-edge GNN architectures, improving task accuracy by 50%+.
- Developed and presented a technical report on a novel GNN architecture, demonstrating significant performance gains and guiding next-phase research.

Leidos Innovation Center
Software Engineering Intern, High-Performance Systems
May 2024 — August 2024 Arlington, VA

- Created and refined GPU-based signal processing algorithms in CUDA, Rust, and Python for real-time applications, improving performance by 400%.
- Developed a real-time machine learning system to lock onto hidden radar targets using PyTorch with 85%+ accuracy.

Peraton
Software Engineering Intern
May 2023 — August 2023 Chantilly, VA

- Automated document processing systems using AI-based parsing systems to increase processing efficiency.
- Authored and presented a whitepaper on systems and processes developed.
- Managed daily intern standups and kept tasks on track.
- Created data ingest and management dashboards for U.S. contractors.

Alluja LLC
Founder
2022 — Present Chantilly, VA

- Creator of multiple products including mobile apps (SwiftUI), libraries (Swift, Rust), backend servers (Rust, Python), and frontends (Svelte, Leptos) impacting 5,000+ users.

Objective

Seeking work in deep reinforcement learning and reasoning: finding new ways to make models learn complex policies, with applications in problem analysis, critical thinking, robotics, vision, and low-data learning.

Education

Purdue University Honors College
Computer Science B.S., Artificial Intelligence B.S.; Minor in Mathematics; Graduated Spring 2026
Fall 2022 — Spring 2026 West Lafayette, IN

- Relevant Coursework: Data Structures, Graduate Machine Learning, Artificial Intelligence, Analysis of Algorithms, Abstract Algebra
- Study Abroad: University College Dublin, Spring 2024

Skills/Exposure

- Python
- Rust
- C
- C++
- C#
- Java
- Kotlin
- Swift
- TypeScript
- JavaScript
- SQL
- PyTorch
- NumPy
- SciPy
- Pandas
- CUDA
- CUTLASS
- Burn
- SvelteKit
- Astro
- React
- Leptos
- Actix Web
- SwiftUI
- Jetpack Compose
- GStreamer

Tools

• EC2 • ECS • S3 • Docker • Docker Compose • MongoDB

- Komori, a Discord Siri Shortcuts integration, with paid downloads on the App Store.
- Alluja WebSockets, a WebSocket testing client for macOS, with 100+ downloads.

Research & Leadership

Purdue Hackers

Engineering Division Lead & Interim President

Jan 2024 — May 2026 West Lafayette, IN

- Oversaw all engineering projects for Purdue Hackers, a 2000+ member student organization focused on innovation and rapid prototyping.
- Managed a team of 10+ officers, balancing budget and time constraints while delivering multiple large-scale technical initiatives.
- Designed and maintained infrastructure for the Passports initiative, including authentication and management systems in Rust and TypeScript, supporting hundreds of active users.
- Led development of The Sign, a meter-tall Conway glider installation representing the club's logo; project was accepted to Hackaday and drew 1,000+ online views.
- Spearheaded the Beacons project in collaboration with Era Labs, deploying early prototypes at student events.
- Organized and ran major events, including a 200+ person callout, hack nights, and technical workshops.

Detecting Source Code Plagiarism in Submitted Assignments

Lead Researcher

Spring 2023 — Fall 2025 West Lafayette, IN

- Selected by Purdue faculty to lead an independent research project addressing plagiarism detection in the Computer Science program.
- Designed a language-agnostic detection system using Abstract Syntax Trees, parsing techniques, and machine learning to identify similarity beyond surface-level code patterns.
- Conducted large-scale testing across multiple courses, targeting 1,000+ student submissions annually to improve fairness and academic integrity.